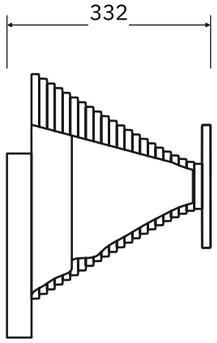
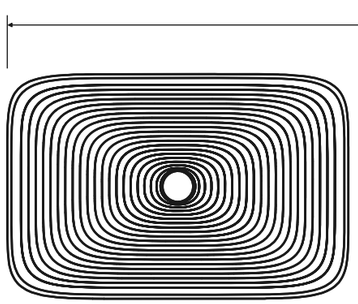
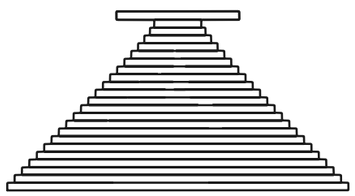
Scale 1:10. First angle: the view from the right is drawn to the left of the front, the view from above under it. A-A: the section on sheet 2.



FROM THE RIGHT

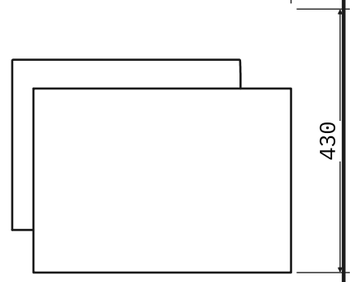


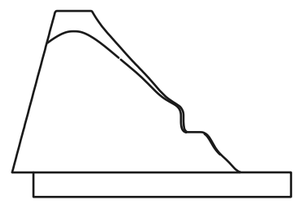
FRONT



FROM ABOVE







General arrangement

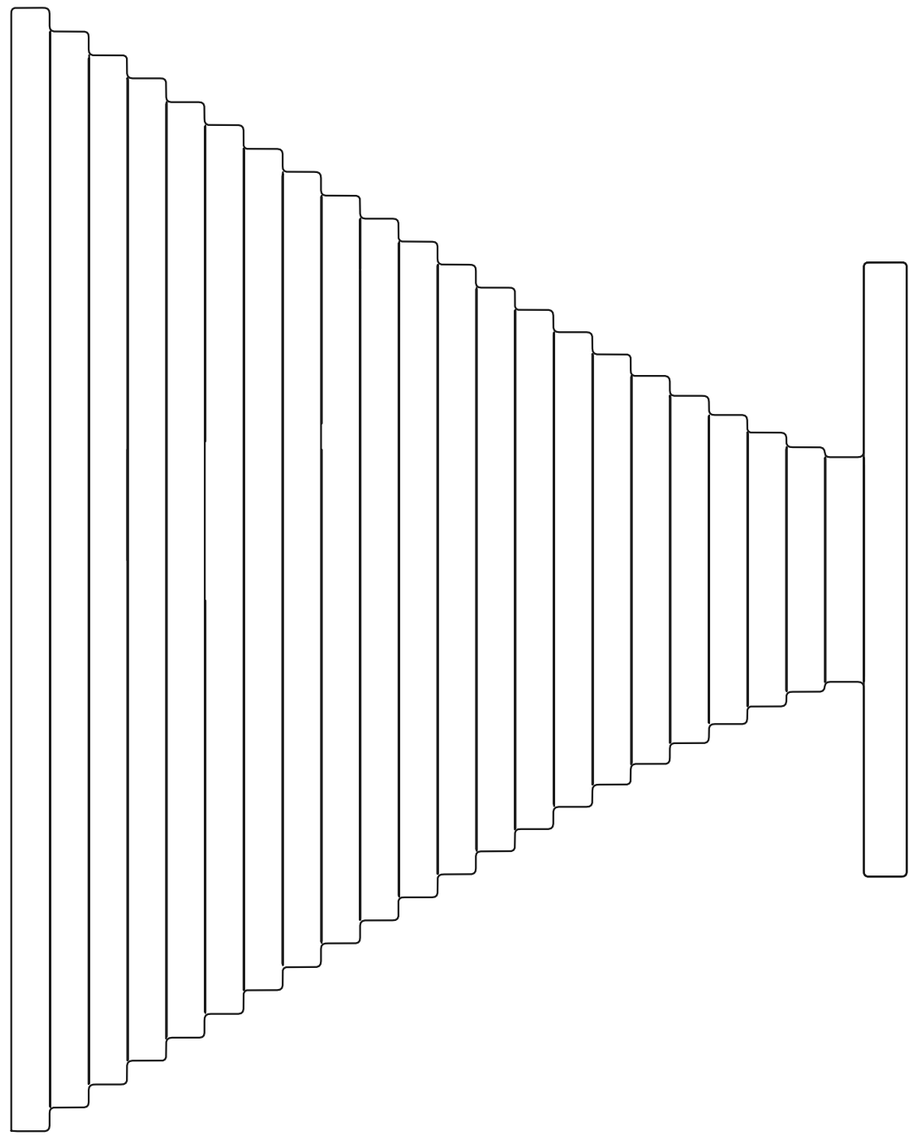
barkhorn: Tropos horn: the forming trial's plug, and Plan B, the laminated horn

mm | issue 2026-11-01 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

BHH-001 rev A

Scale 1:2. Section A-A (sheet 1): the product cut at x 1517.9 and seen from the right; cut faces hatched, a different angle for each part.



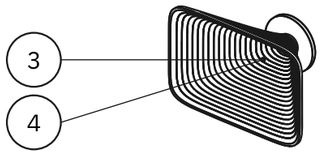
Section A-A

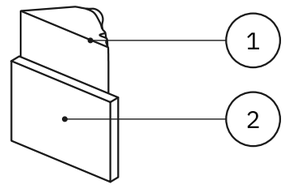
barkhorn: Tropos horn: the forming trial's plug, and Plan B, the laminated horn

mm | issue 2026-11-01 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:20, parts drawn apart. Balloons are the parts list's numbers (last sheet).





Exploded view

barkhorn: Tropos horn: the forming trial's plug, and Plan B, the laminated horn

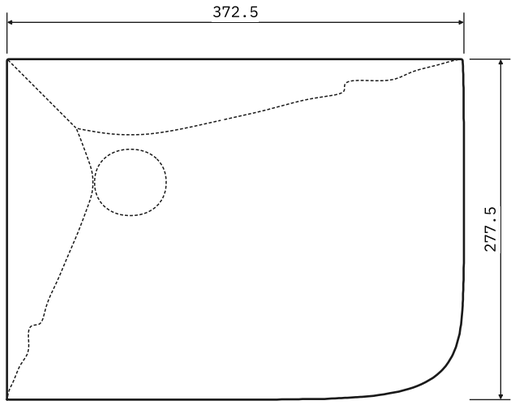
mm | issue 2026-11-01 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

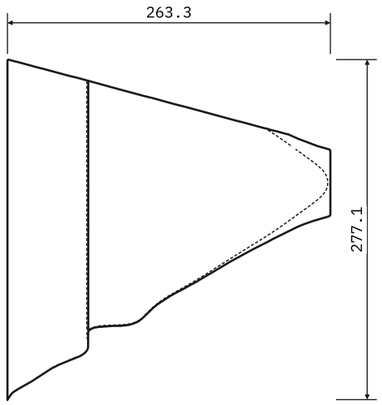
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

1. horn-trial-plug: cnc-mill, tooling-board; 10031 cm3 each; 1.0 for the set; 1:5

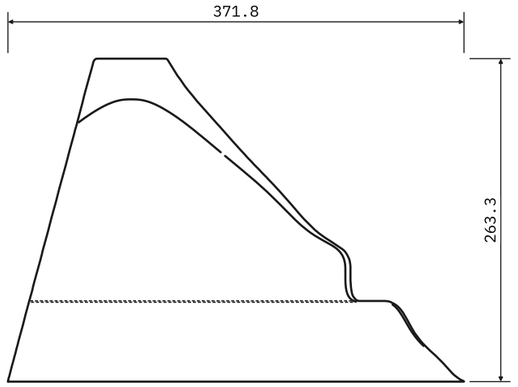
the forming trial (make/forming-trial.md): the production plug's surface cut to one quadrant, the throat end and the mouth's lower corner at full scale, its cut faces drafted 15 degrees outward toward the base and 30 past the axis; the same 5-axis program as the full plug, in the same board; formed in stock 10 mm cast sheet, then gauged for wall and colour before the full plug is cut



FRONT



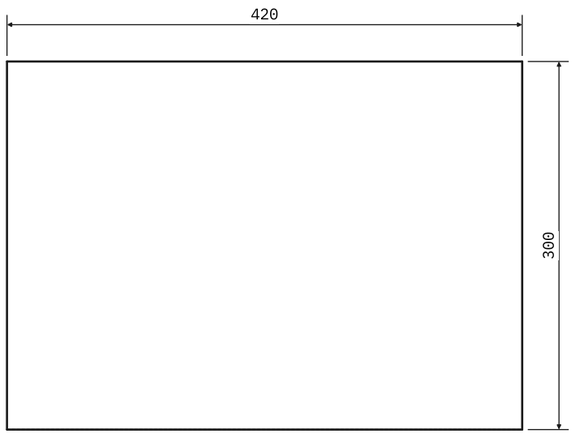
FROM THE RIGHT



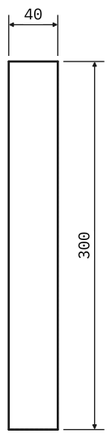
FROM ABOVE

2. horn-trial-base: cnc-mill, tooling-board; 5040 cm3 each; 1.0 for the set; 1:5

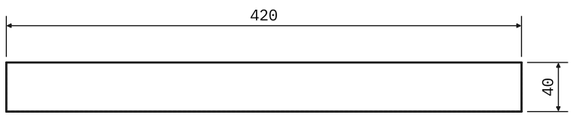
the trial plug's base, in board, bolted to the forming machine's table; the clamp frame sized to it



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

barkhorn: Tropos horn: the forming trial's plug, and Plan B, the laminated horn

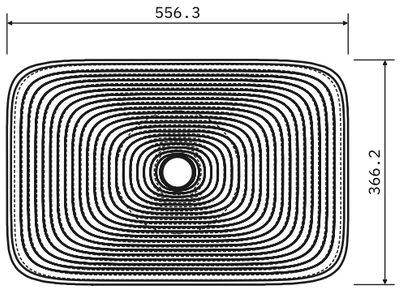
mm | issue 2026-11-01 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

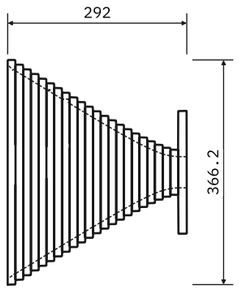
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

3. horn-plan-b: cnc-mill, stock-cast-sheet; 3514 cm3 each; 2 for the set; 1:10

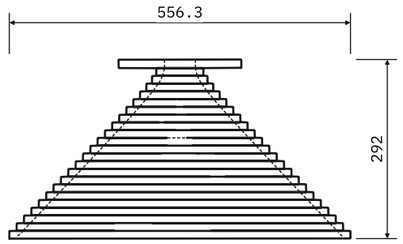
Plan B (make/plan-b.md, make/laminated-horn.md): the flare and the neck laminated from 22 layers of stock 12.7 mm (1/2 in) tinted cell-cast sheet, each layer faced to its pitch, and the boss; each layer CNC-cut to its outline (10 outside the air path), bonded with polymerising cement in a jig, annealed, and the air path machined through on 5 axes and polished smooth (the bond lines cross it flush); the outside stepped, a terrace a layer, each layer's edge polished; in place of the formed flare and the throat piece, on the same pod, cap, pads and posts



FRONT



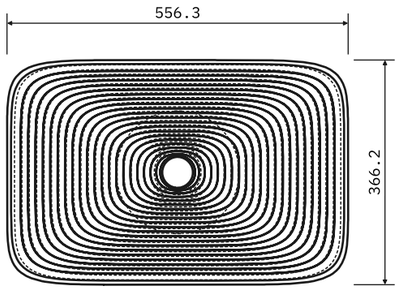
FROM THE RIGHT



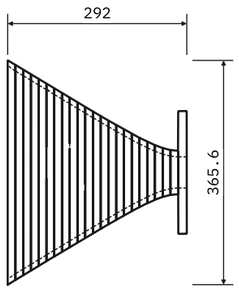
FROM ABOVE

4. horn-plan-b-smooth: cnc-mill, stock-cast-sheet; 2310 cm3 each; 2 for the set; 1:10

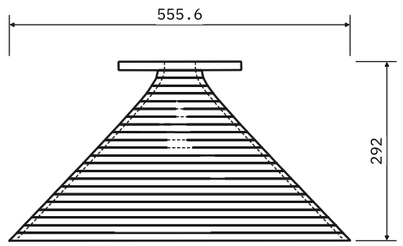
Plan B (make/plan-b.md, make/laminated-horn.md): the flare and the neck laminated from 22 layers of stock 12.7 mm (1/2 in) tinted cell-cast sheet, each layer faced to its pitch, and the boss; each layer CNC-cut to its outline (10 outside the air path), bonded with polymerising cement in a jig, annealed, and the air path machined through on 5 axes and polished smooth (the bond lines cross it flush); the outside machined smooth on 5 axes to follow the flare, then polished; the bond lines run round it as fine rings; in place of the formed flare and the throat piece, on the same pod, cap, ...



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

barkhorn: Tropos horn: the forming trial's plug, and Plan B, the laminated horn

mm | issue 2026-11-01 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

For 2; prices are budgets (bom.md says where each comes from).

no.	item	make	qty	what	size	each USD
1	horn-trial-plug	cnc-mill	1.0	High-temperature epoxy tooling board, glued up as a block, CNC-cut, sealed (the prototype's forming plug; lead time 3 to 4 weeks from the program)	382 × 273 × 364	2,054.80
2	horn-trial-base	cnc-mill	1.0	High-temperature epoxy tooling board, glued up as a block, CNC-cut, sealed (the prototype's forming plug; lead time 3 to 4 weeks from the program)	430 × 50 × 310	321.63
3	horn-plan-b	cnc-mill	2	Stock cell-cast acrylic sheet, 1/2 in (12.7 mm), tinted through in a standard colour (2423 Red for Ember; 2064 or 2074 Gray for Smoke), bought by the 48 × 96 sheet with no minimum (AC Plastics, 2423 at $835.80 a sheet, 2026-10-31; research	1000 × 1713 × 13	1,482.62
4	horn-plan-b-smooth	cnc-mill	2	Stock cell-cast acrylic sheet, 1/2 in (12.7 mm), tinted through in a standard colour (2423 Red for Ember; 2064 or 2074 Gray for Smoke), bought by the 48 × 96 sheet with no minimum (AC Plastics, 2423 at $835.80 a sheet, 2026-10-31; research	1000 × 1713 × 13	1,707.62

Notes

1. The forming trial (make/forming-trial.md): the production plug's surface cut to one quadrant at full scale, the throat end and the mouth's lower corner, its cut faces drafted outward; in high-temperature epoxy tooling board, on a board base; formed in stock 10 mm cast sheet and gauged before the full plug is cut. One set (the parts list counts it once).
2. Plan B (make/plan-b.md, make/laminated-horn.md): the flare and the neck laminated from 22 layers of stock 1/2 in tinted cell-cast sheet, each layer CNC-cut to its outline, bonded with polymerising cement in a jig, annealed, the air path machined through on 5 axes and polished; the outside stepped (horn-plan-b) or machined smooth (horn-plan-b-smooth). Each stands in the formed horn's place on the same pod, cap, pads and posts; the parts list counts both, one is chosen.

Parts list and notes

barkhorn: Tropos horn: the forming trial's plug, and Plan B, the laminated horn

mm | issue 2026-11-01 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15